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Kinetic analysis of Ni—Co electrodeposition with saccharin using electrochemical impedance spectroscopy

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ABSTRACT

The reaction kinetics of nickel-cobalt alloy electrodeposition is investigated through Electrochemical Impedance Spectroscopy (EIS). Results revealed two primary effects of saccharin additives: reducing competition between hydrogen and metal adsorption, and inhibiting reaction by blocking active sites. At low saccharin concentrations, reduced competition led to rapid hydrogen evolution, resulting in non-uniform and rough microstructures. Conversely, saccharin's strong inhibition effect at higher concentrations yielded fine microstructures with mirror-like appearance. These microstructural changes significantly enhanced the hardness and reduced friction of the electrodeposited film. Furthermore, the suppression of hydrogen reactions stabilized the local pH during the electrodeposition, producing high-quality metallic films with reduced oxide/hydroxide impurities. This study paves the way to understand the complex interplay of electroplating additives in metallic-alloy electrodeposition to optimize microelectronics fabrication.

1. Introduction

Micro-electro-mechanical systems (MEMS) probe cards are essential instruments in the semiconductor manufacturing process that facilitate electrical testing of integrated circuits (ICs) on a semiconductor wafer through the electrical die-sort (EDS) testing technique. Because of the ongoing scaling of the semiconductor devices, wafer reliability testing has become extremely demanding and critical requiring sophisticated material optimization. The probe card comprised of microscale probe arrays serves as the core component of this test enabling intricate positioning and current flow to the ICs to sort out wafers with defects.

The low strength and hardness of the probe tips in high temperatures have become an issue because of the harsh conditions met during the testing. This comes from the advancement in semiconductor scaling that necessitates the downsizing of probe tips, concentrating the heat generated from the current flow. The probes are very fragile—typically a millimeter long and 50 μm thick—when vertically applying force to the wafer while passing current to analyze defects. Therefore, maintaining high-temperature strength and electrical conductivity during the inspection process is crucial, given that the temperature rises nearing 200 °C. Furthermore, minimizing deformation in the contact area and maintaining optimal tribological properties are necessary to prevent damage to the wafer surface.

Nickel and cobalt are widely recognized for their exceptional mechanical, electrical, and tribological properties and are extensively utilized across various industrial sectors [1–8]. However, mixing these metals to form Ni—Co alloys is even more interesting because of the widely tunable material properties. These alloys that form solid solutions in the entire composition range [9–12], have emerged as compelling alternatives to expensive metals due to their wide range of tunable compositions rendering tunable properties required by various applications [13–21]. Their unique tunable properties and relatively low cost make them particularly attractive in applications where both performance and cost efficiency are critical, for instance, the microelectronics and semiconductor industries [22–25]. As a result, Ni—Co alloys have also gained large attention for their application in fabricating MEMS probe cards due to their high-strength, and tunable hardness, tribological properties [26–28].

Among the specified requirements, the probe tip's strength and conductivity predominantly hinge on its material composition and

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structural design. In this work, we employ a Ni-Co/Cu multilayer configuration of MEMS probe tip with a Ni, Co composition ratio near 70:30. Previous reports have addressed the strength and conductivity criteria by systematically analyzing the stacking sequences and component filling ratios that are applied in the structural design [29]. Nonetheless, the reliability of the probe tip remains a concern, primarily attributed to the low hardness and elevated friction of the contact area. Although the structure's strength and conductivity largely rely on the material composition and structural design, the surface property has to be intricately engineered by optimizing the film's microstructure, posing significant challenges to fabricating the probe cards. This challenge can be addressed by engineering the microstructure through grain refinement and controlling the impurities, achieved in this work by optimizing the reaction kinetics [30].

There are several difficulties in controlling the growth of Ni—Co films during electrodeposition such as the unique mechanism during the electrodeposition. Electrodeposition emerged as a preferred method for fabricating Ni—Co films for the probe card because of its simplicity, cost-effectiveness, and compatibility with the micro-patterning process. However, a unique deposition mechanism of nickel and cobalt complicates the process, which is the anomalous co-deposition, a mechanism in which the deposition rate of less noble metal is higher than that of noble metal, contrary to the expected behavior. This results in the incorporation of undesirable impurities such as metal oxides and hydroxides, compromising film quality and mechanical properties. It has been reported that anomalous co-deposition occurs because the reduction potential of nickel and cobalt changes by forming an intermediate species combined with hydroxide ions during the electrodeposition [31].

After the formation of intermediate species, it covers the electrode where the deposition is occurring and could be embedded inside the film. The intermediate formation is typically known to be especially strong when the electroplating solution is non-acidic. However, it has been numerously reported that it occurs even in an acidic solution when the local pH rises significantly. For instance, hydrogen gas generation in acidic baths, which is the prominent side reaction, induces a huge local pH elevation leading to a significant increase in the precipitation of metal hydroxides [32–37]. Thus, meticulous control over the reaction is essential in enhancing the film quality.

Using electrodeposition additives is a simple and cost-effective method to effectively address these challenges. A proper selection of electrodeposition additives, through its interaction with the electrode, enables control of nucleation and growth of electrodeposited films [38,39]. The additives can be selected based on the preferred target properties such as brightening, suppressing, leveling, etc. [40–45] However, to our knowledge, detailed research on the use of additives for the kinetic control of multiple reactions remains unexplored, and studies investigating their underlying mechanisms are still lacking. Therefore, to address the critical gap in unveiling the mechanisms of electroplating additives and practical usage for the fabrication process, we utilize saccharin to study its effect on the reaction kinetics and aim to elucidate the mechanism of growth control. Through an approach encompassing theoretical analysis aided by experimental characterization, we propose a means of carefully controlling specific reactions in the Ni—Co electrodeposition process using semi-empirical electrochemical impedance spectroscopy (EIS) analysis. We utilize EIS as the primary method for studying the kinetics of Ni—Co electrodeposition with saccharin due to its ability to simultaneously evaluate reaction kinetics and steady-state surface coverage. While alternative techniques such as cyclic voltammetry (CV) and chronoamperometry offer valuable kinetic insights, EIS provides a relatively simple and efficient approach in assessing complex surface interactions, charge transfer resistance, and adsorption behavior of electrochemical species. Furthermore, by incorporating theoretical derivations, this approach enhances the quantitative interpretation of EIS data, making it a powerful tool for understanding the effect of additives in electrodeposition and revealing it's mechanism.

A brief overview of saccharin as an electroplating additive is presented here to aid the understanding of its role in modifying electrochemical reactions. Saccharin is one of the organic additives often used in electroplating. The molecular formula of saccharin is $C_7H_5NO_3S$. It has S and O atoms, and carbonyl groups, which have lone pair electrons and π electrons. This gives saccharin a distinct ability to interact with the electrode during the electrochemical process, resulting in various advantages to deposited films such as improving mechanical properties and brightness [46,47]. According to previous studies, saccharin is primarily known to act as an inhibitor in Ni electrodeposition [48], and change the Ni(II) reduction kinetics by decreasing $Ni_{ad}(I)$ [49]. However, despite the benefits of saccharin on film properties, it can leave some sulfur and carbon impurities in coatings which can cause embrittlement and poor corrosion properties [47,50,51]. As a result, proper amounts of saccharin must be added to plating solution to have desired properties with minimal side effects. Furthermore, saccharin has been reported to regulate anomalous co-deposition in Ni—Fe and Fe—Co alloys, enabling enhanced control of the alloy composition [52,53]. However, while saccharin has been suggested to suppress the side reactions such hydrogen evolution, direct experimental evidence remains lacking, to our knowledge [53]. Our findings in this paper not only corroborate previous studies but also provide novel insights into saccharin's role in electrodeposition. Specifically, we elucidate its influence on suppressing hydrogen evolution, alteration of saccharin's function at low concentrations, and its effect on crystal orientation, as well as demonstrating significant improvements in hardness and friction. Unlike prior studies that primarily attribute saccharin's effect to general growth inhibition, our kinetic analysis identifies hydrogen suppression as a key mechanism governing microstructural evolution and property enahcement. These discoveries offer new insights into engineering surface properties and improving microelectronic fabrication.

2. Experimental

2.1. Electrolyte and electrodes

A copper electrode of $1cm^2$ was used as the cathode. On a glass substrate, 300 nm of copper was deposited on top of a titanium adhesion layer of 30 nm through the DC sputter system. All copper electrodes were cleaned in order of ethanol, isopropyl alcohol, and DI water for 3 min with ultrasonic treatments. All measurements were performed with a platinum sheet with a surface area of $6.25cm^2$ and a thickness of 0.25 cm as the counter electrode, and a standard calomel electrode (SCE) as a reference electrode. All chemical substances used are analytically graded. Ni—Co electroplating baths were prepared according to the following content. Boric acid was added to stabilize the solution pH to 4, and four different concentrations of saccharin were added to the solution, 0.05, 0.3, 0.5, and 1 g/L, respectively. For the electrochemical characterization (linear sweep voltammetry (LSV), and EIS) and film deposition for surface/mechanical characterization, 200 mL solution was used, whereas for the MEMS probe tip fabrication, 1 L solution was used. Moreover, all electrodepositions were carried out in an applied current density of 2 A/dm^2 for 20 min. For convenience, each of the electrodeposited specimens from each solution is referred to as A, B, C, D, and E as in the table.

2.2. Measurement equipment and methods

Linear sweep voltammetry (LSV) measurement was performed by applying a voltage from open circuit potential (OCP) to −1.5 V with a scan rate of 5 mV/s, which was confirmed to be stable. EIS measurements were performed with an amplitude perturbation of 5 mV at an overpotential of 0.4 V in the negative voltage. Through changing the perturbation, 5 mV was also found to be sensitive and stable enough to capture the dynamics in the electrochemical double layer with a low signal-to-noise ratio. All electrochemical measurements were performed

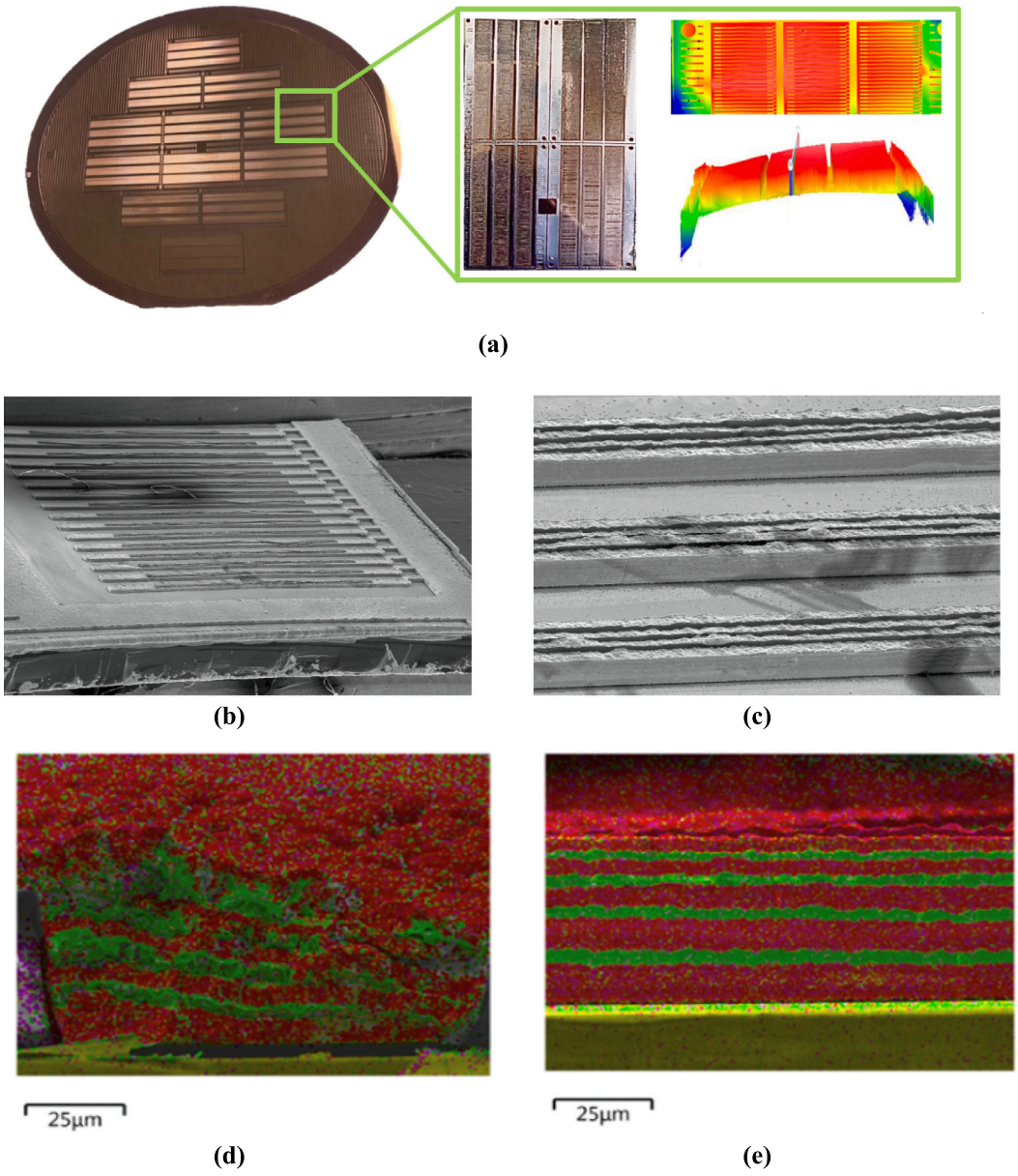


Fig. 1. (a) Patterned wafer used for MEMS probe tip fabrication. The electrodeposited film stacks are polished on both the top and bottom sides using chemical mechanical polishing (CMP) after electrodeposition to ensure proper dimension. A 500 nm PVD copper seed layer is deposited on a Si wafer with a Ti adhesion layer. The right box presents a zoomed-in image of the patterned wafer after electrodeposition, along with 3D optical scanning images before electrodeposition. The red-orange regions indicate areas of greater thickness, corresponding to the placement of a negative photoresist for pattern formation. (b) SEM images of electrodeposited probe tip structures consisting of seven layers of Ni-Co/Cu, with a total thickness of 50 μm. (c) A magnified view of (b), highlighting the pattern details. (d, e) Energy-dispersive X-ray spectroscopy (EDS) mapping images demonstrating the improvement in multilayer structure uniformity due to saccharin addition. The structure transitions from a non-uniform morphology (d) to a uniform morphology (e). The red, green, and yellow elements correspond to detected Ni, Cu, and Si, respectively. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

by a VSP Potentiostat from Biologic. Biologics' EC-Lab software was used to fit the experimented data of EIS measurements. To examine the surface morphology of the deposited films, a JSM-7600F Scanning Electron Microscope (SEM) from Jeol was used. The surface microstructure was imaged by a magnification of 30,000 with an accelerating voltage of 15 kV. A SmartLab X-ray Photoelectron Spectroscopy (XPS) from Rigaku was used to characterize the surface chemical bond energy. Before measuring chemical states, the surface was etched with argon ion plasma for 10 min to remove surface oxide films. For the measurement of mechanical and tribological properties, nano hardness and micro scratch tests were performed. For these tests, a CSM NST nano-indenter and MST-CSEMEX micro scratch tester were employed. Hardness measurement was repeated nine times for each specimen by indentation loads of 50mN with a loading and unloading rate of 100mN/min. Mean values of measurements with standard deviations were used to indicate hardness and Young's modulus values. Scratch tests were performed with a progressive load of 30mN to 3000mN and a scan rate of 2970mN/min. Measurements were carried out for 1 min with a scratch distance set to 0.5 mm.

3. Results

3.1. Enhancing microelectronic fabrication with saccharin

The excellent pattern formation capability of our electroplating solution with saccharin is demonstrated and presented in Fig. 1. Decreased

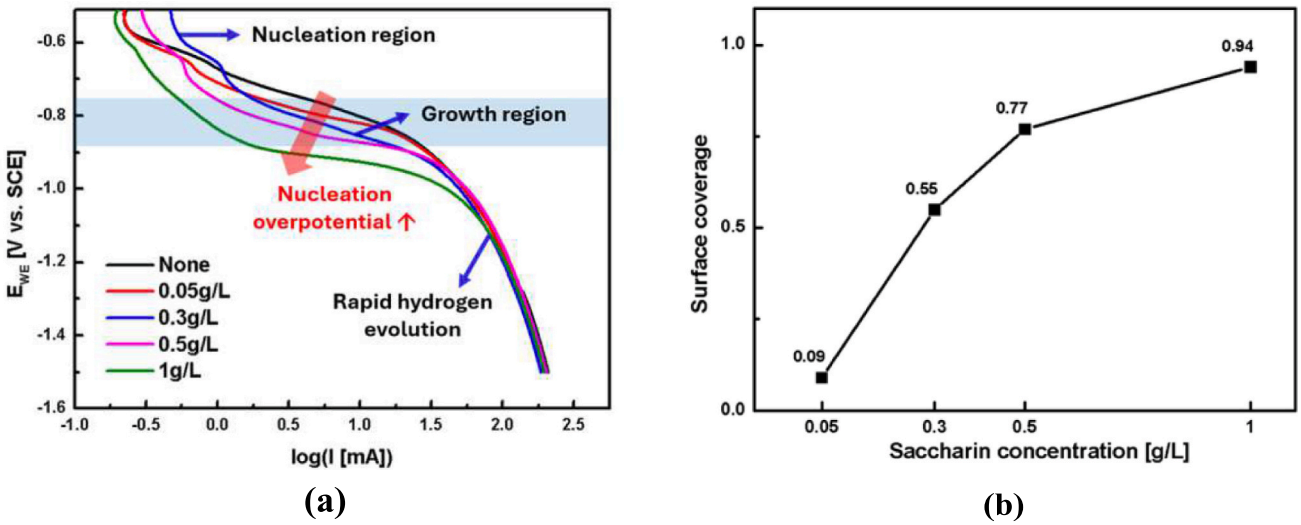


Fig. 2. (a) Linear sweep voltammetry (LSV) curve for different saccharin concentrations measured at a scan rate of 5 mV/s, presented on a logarithmic scale. The regions of nucleation, growth (blue-shaded), and rapid hydrogen evolution are indicated for the 0.3 g/L saccharin case. The shift of the growth region to higher overpotentials indicates an increase in nucleation overpotential. (b) The surface coverage of saccharin on the cathode at different concentrations determined by the Langmuir adsorption isotherm and measured data from LSV. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

film impurities and controlled electrochemical reactions lead to a uniform film with enhanced hardness and decreased friction. As seen in the zoomed-in image in the top right, which exhibits the appearance after the 7 layer Ni-Co/Cu multilayer stack electrodeposition, the films have a bright appearance without dims. The optical 3D scan images on the right of Fig. 1(a) shows the negative photoresist (PR) pattern before electrodeposition with the red-orange region indicating greater thicknesses. The PR thickness was measured to be 50 μm. The SEM image in Fig. 1(b, c) confirms excellent micro-pattern formability. Lastly, Fig. 1(d, e) captures the cross-section image of the improved film uniformity with saccharin addition. A high aspect ratio trench can be filled uniformly without voids largely because of the impurity-free, compact growth rendered by saccharin. However, when the trench dimension is reduced to deeply sub-microns, additional suppressor additives such as high molecular weight polymers are needed in conjunction with saccharin for achieving void-free trench filling. Saccharin also shows good compatibility with other additives such as suppressors and levelers, providing a large potential in microelectronic fabrication.

3.2. Electrochemical, surface characterization

3.2.1. Linear polarization measurement

Linear Sweep Voltammetry (LSV) measurements were carried out in the cathodic polarization potential. This measurement sweeps the applied voltage with a constant scan rate and gives characteristic information about the reactions, such as the peak positions, Tafel slope, and exchange current density. After testing various scan rates, 5 mV/s was determined as optimal for this experiment having an optimal balance between measurement time and signal quality. A fast sweep obscured data interpretation because the chemical components in the solution lack sufficient time to stabilize, leading to accumulated current measurements. Conversely, a slow scan rate introduced noise due to temperature fluctuations and extend measurement time. Notably, the resulting deposits after the LSV measurement had different appearances. Gray deposits were observed from the None/0.05 g/L concentrations, while metal-silver deposits were obtained at higher concentrations over 0.3 g/L. During the electrodeposition of Ni—Co alloy, multiple reactions occur simultaneously, including diffusion on the cathode surface, nucleation, growth of Ni and Co, hydrogen evolution, etc. Each reaction contributes a distinct Tafel slope, causing the slope to change over the overpotential. To better observe changes in the low overpotential region, important for studying the nucleation process, log-scaling the current is useful.

In Fig. 2(a), different Tafel slopes appear across different overpotential ranges, reflecting the dominant electrochemical processes at each stage. During the nucleation process of the Ni—Co electrodeposition, the coalescence of metal atoms through diffusion occurs which transitions to continuous growth once the nuclei reach the critical size. The stable growth of Ni—Co alloy occurs at a region where the current changes follow a linear slope over a wide range of current, after the nucleation region. In this region, a small overpotential change accompanies a large current increase rendering it suitable for controlling the deposition rate. The corresponding region for the case of 0.3 g/L is presented in Fig. 2(a) as the blue-shaded region. The region with steeper slopes and peaks in the lower overpotential (−0.5 V to −0.75 V) is where the nucleation process is prevalent while hydrogen evolution dominates in the high overpotential range where the slope becomes much steeper.

The characteristic Tafel slope in the growth region differs by the additive concentration as shown in Fig. 2(a). At concentrations of 0.3 g/L and above, the slope is steeper than at 0.05 g/L or without saccharin. This suggests that saccharin affects the nucleation and growth rates of Ni and Co differently at high and low concentrations [54]. The Tafel equation (Eq. 1) is extrapolated from the LSV curve for further analysis, and additional parameters are obtained from the Tafel equations such as the exchange current density, and charge transfer coefficient. The exchange current density (i_0), defined in Eq. 2, represents the rate of the forward and reverse electrochemical reactions at equilibrium, where the net current flow is zero due to equal and opposite reaction rates of the equilibrium state with no overpotential [55]. The charge transfer coefficient, a symmetry factor of the charge transfer energy barrier (ranging between 0 and 1), is calculated using Eq. 3, where b is the Tafel slope, R and F are the gas constant, and Faraday's constant, respectively [54,55]. The Tafel equations, along with the exchange current density and charge transfer coefficient at various additive concentrations, are summarized in Table 2 using values extracted from Eqs. 1–3.

$$\eta = a + b\log(i) \tag{1}$$

$$\log i_0 = -\frac{a}{b} \tag{2}$$

Table 1
Composition of nickel cobalt electroplating bath.

$Ni(SO_3NH_2)_2 \cdot 4H_2O$	300 g/L				
$NiCl_2 \cdot 6H_2O$	45 g/L				
$Co(SO_3NH_2)_2 \cdot xH_2O$	50 g/L				
H_3BO_3	30 g/L				
Saccharin	None	0.05 g/L	0.3 g/L	0.5 g/L	1 g/L
Deposited samples	A	B	C	D	E

$$\alpha = RT/bnF \tag{3}$$

$$i_{net} = i_0 \left[e^{-\alpha \frac{F\eta}{RT}} - e^{\frac{(1-\alpha)F\eta}{RT}} \right] \tag{4}$$

Saccharin tends to suppress the electrochemical reaction as indicated by the extracted parameters in Table 2. For instance, the i_0 values tend to decrease with the addition of saccharin, which generally signifies a reduced electrochemical reaction rate [54]. This is because saccharin additives easily adsorb to the metal by forming a chemical bond and blocking the active sites. It is noteworthy that the exchange current density is almost negligible in 1 g/L compared to other concentrations. Given the distinctly low current density, it is anticipated that the adsorption of saccharin reaches the saturation stage, blocking most active sites where the electrochemical reaction occurs [55].

The suppression of saccharin accompanies a decrease in the net current of the electrochemical system that is described by the Butler-Volmer equation in Eq. 4 [56]. The equation signifies two competing mechanisms on the reaction rate. When overpotential is negative, the net current (i_{net}) increases exponentially with higher α. However, the net current is determined by multiplying the exponential term with the i_0. Although α tends to increase with higher saccharin concentration, the significant drop in the i_0 value results in a decrease in the net current. Therefore, as shown in the red arrow of Fig. 2 (a), the linear growth region shifts to a larger overpotential with higher saccharin concentration, signifying a larger potential energy required for growth. This indicates an increase in nucleation overpotential, defined as the additional potential required to overcome the energy barrier for nucleation and initiate growth (measured by the potential difference between the OCP and the point at which the current increases rapidly). This increase occurs because saccharin blocks active sites, leaving energetically unfavorable sites for metal reactants to adsorb and exchange electrons. Higher nucleation overpotential increases the nucleation rate, as described by $A = A_0 exp\left(-\frac{b}{\eta^2}\right)$, (A is the nucleation rate, η the overpotential, and A_0 and b are constants related to the electrode and electrolyte properties). Consequently, the higher nucleation rate results in reduced grain size, enhancing the hardness and strength of the films [55,57,58,59].

In summary, saccharin inhibits electrochemical reactions and increases nucleation overpotential, followed by an increased nucleation rate and altered surface properties. However, an exception to this general trend is observed at 0.05 g/L. At this low concentration, the increase in α is larger compared to 0.3 g/L (20 % increase for 0.05 g/L and 7 % for 0.3 g/L), contrary to the general trend of larger α at higher saccharin concentrations. The deviation is attributed to increased hydrogen evolution reactions which must be considered alongside metal reductions. The accelerating effects at low saccharin will be further analyzed in the Discussion section by theoretically calculating the kinetics of each electrochemical reaction involved in Ni—Co electrodeposition.

Lastly, the surface coverage of saccharin on the electrode (θ) is estimated by Eq. 5 and presented in Fig. 2 (b), where i_{add} and i are the reduction current values with and without saccharin, determined in the LSV curve (Fig. 2(a)) at the overpotential −0.85 V. This point represents a stable reduction region with a well-defined Tafel slope over a wide overpotential range, effectively reflecting the surface coverage effects of saccharin. Lower overpotential is unstable due to local, inhomogeneous reactions and obscure the analysis, while in the higher overpotential region, the hydrogen evolution reaction dominates. As shown in Fig. 2 (b), surface coverage increases with higher saccharin concentration and saturates when the concentration is high.

$$\theta = \left(1 - \frac{i_{add}}{i}\right) \tag{5}$$

3.2.2. Microstructure and chemical analysis

To examine the surface morphology of the samples deposited from each solution, a scanning electron microscope (SEM) was used to image the surface at 30,000× magnifications. Distinct morphological differences were observed for specimens deposited from low and high saccharin concentrations. Specimens are labeled as in Table 1: Specimen A for the deposit without saccharin, and Specimen B, C, D, and E for those deposited with progressively higher saccharin concentrations. In specimens A and B uneven surfaces of pointed shapes were formed as shown in Fig. 3 (a, b). However, the surface morphology changed dramatically to compact and uniform surfaces in specimens C, D, and E as seen in Fig. 3 (c-e). As analyzed above, the dramatic change in surface morphology is partially attributed to saccharin's role in promoting nucleation through increased overpotential, resulting in the formation of finer grains and compact/uniform microstructures. However, controlled hydrogen reactions and reduced intermetallic impurities have a substantial effect on uniform growth, as will be discussed in later sections.

While grain size reduction contributes to improved uniformity, the significant chemical variations identified by the XPS analysis of the films suggest an additional underlying factor. Notably, the grain size extracted from the XRD analysis, showing approximately a 34 % reduction in grain size in 0.3 g/L (in Fig. 4(b)), confirms that the grain size reduction is insufficient to fully explain the observed differences in morphology. Therefore, the XPS spectra of the surface of deposited films were obtained for detailed chemical analysis after removing the surface oxide films through the Ar ion sputtering process. Four peaks are observed in the Ni2p scan for all samples. The relative intensity of the two shake-up peaks differs significantly between the lower and upper curves in Fig. 3 (f), which represent lower/without and higher saccharin concentrations, respectively. In specimens A and B, the two shake-ups of $Ni2p_{2/3}$ and $Ni2p_{1/2}$ were observed near 880 eV and 862 eV in which the intensities are comparable to the main peaks [60]. In specimens C, D, and E, the peak intensities of the two shake-ups are significantly smaller compared to the $Ni2p_{2/3}$ and $Ni2p_{1/2}$ peaks located near 853 eV and 870 eV. This spectrum is characteristic of metallic Ni, indicating specimens C, D, and E predominantly contain Ni in their metallic state [61]. In contrast, for specimens A and B, the $Ni2p_{2/3}$ peak aligns well with the binding energies of 855-856 eV, corresponding to the Ni^{2+} and Ni^{3+} peaks [61]. This suggests that significant amounts of nickel hydroxides and oxides are present in these films. This inference is further supported by the high O 1 s signal observed in specimens A and B in Fig. 3 (h) [62,63].

The Co peak exhibits similar behavior, indicating that specimens C, D, and E primarily contain metallic Co, whereas specimens A and B contain cobalt hydroxides and oxides. The $Co2p_{2/3}$ peak is located at 778 eV for specimens C, D, and E as shown in Fig. 3 (g). These correspond to the Co metal peaks [64]. For specimens A and B, peaks are shifted to the left, 780 and 781 eV, respectively [64]. These correspond to Co^{2+} and Co^{3+} peaks, which are present in $Co(OH)_2$ and CoO(OH). In Fig. 3 (h), compared to specimens A and B, the O1s peak intensities are much smaller for specimens C, D, and E showing much less oxygen content. In general, metal-hydroxides have a binding energy of 531-532 eV, and metal-oxide bonds have an energy of 528-531 eV which is consistent with the peaks of specimens A and B indicating large presence of oxide/hydroxide impurities [64,65].

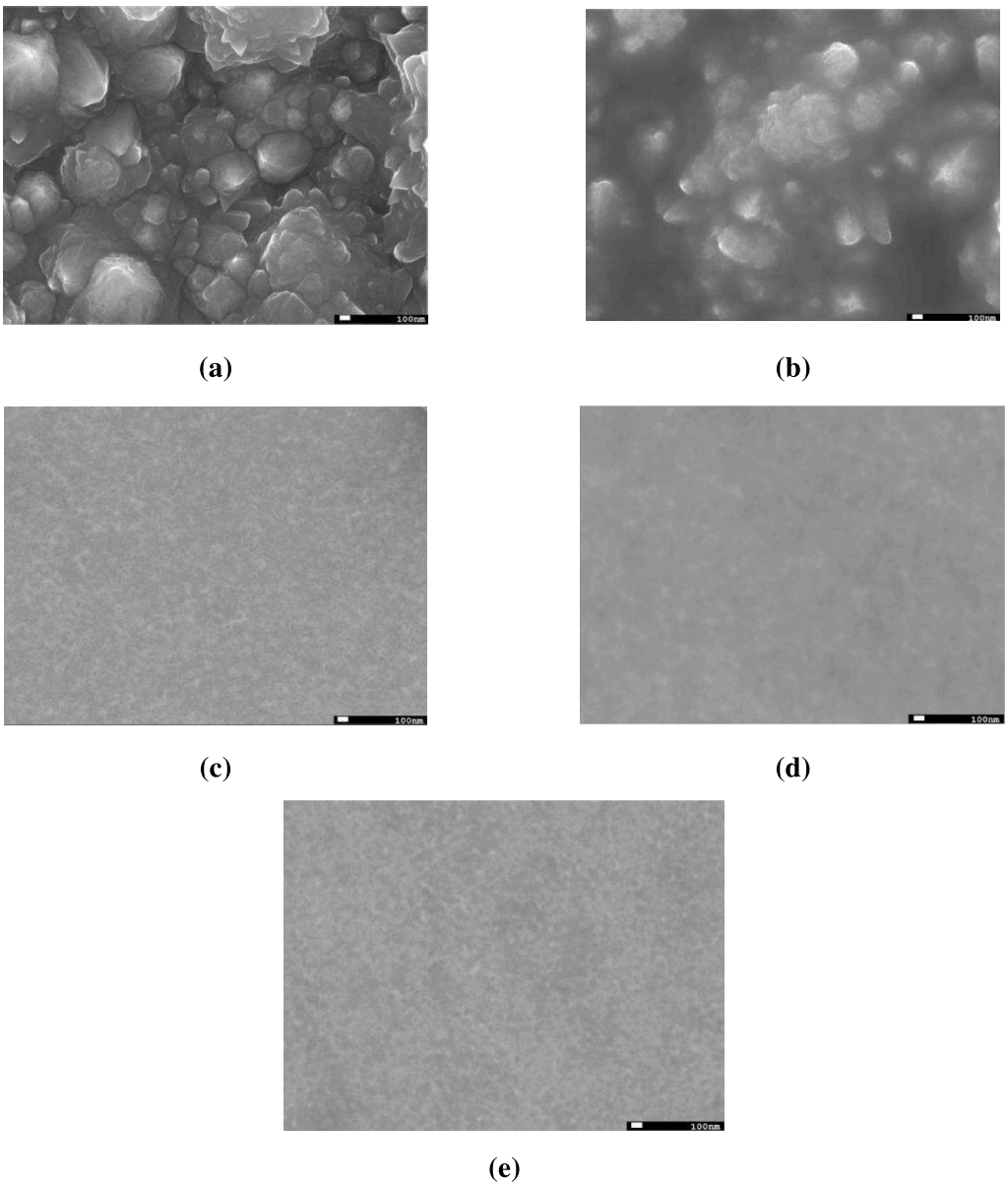


Fig. 3. SEM image (30,000× magnification) of Ni—Co deposits from different saccharin concentrations. (a) Surface morphology of deposit without saccharin (b) film deposited in 0.05 g/L, (c) 0.3 g/L, (d) 0.5 g/L, and (e) 1.0 g/L. (f - h) are the high-resolution XPS spectra of the films deposited at different saccharin concentrations; Ni2p scan, Co2p scan, and O1s scan respectively.

3.2.3. *Hardness and friction coefficient*

The mechanical and tribological properties of the Ni—Co film are significantly improved by the addition of saccharin due to the altered grain size and morphology. First, Vickers' hardness of the specimens was measured through nano-indentation tests. The results are shown in Fig. 4 (a). Second, Young's modulus was obtained, as shown next to the hardness values in Fig. 4 (a), through the identical nano-indentation setup by measuring the gradients of the stress-strain curves when unloading the indent. Adjusting the proper indentation depth is significant to obtain an accurate result when measuring the hardness/stiffness value of films, contrary to the cases of bulk material which is insensitive to the depth. An indentation depth of less than 10 % of the film thickness could result in incorrect measurements by the surface roughness effects [66]. On the other hand, indentation depth exceeding 17.5 % can also give biased values due to the substrate effect [66]. Therefore, all measurements were performed with controlled penetration depth between 10 and 15 % of the entire coating thickness which was within 7-8 μm [66]. Compared to specimens A and B, the hardness values of specimens C, D, and E increased by more than 50 %, attributed to the compact microstructure and reduced grain size. Additionally, the small error bars of the hardness values in specimens C, D, and E (Fig. 4 (a)) reflect the improved uniformity of the films.

The difference in hardness among high-saccharin concentration films (above 0.3 g/L) is relatively small but follows an interesting trend. In these films with compact surface morphology, grain size becomes the dominant factor in determining hardness. According to the Inverse Hall-Petch relation, in nanocrystalline films with grain sizes below 10 nm, further grain size reduction leads to a decrease in hardness [67–70]. This effect identifies 0.3 g/L as the optimal concentration for achieving a hard coating with well-balanced mechanical properties. The grain size reduction, inferred from the increased nucleation rate, is confirmed by the grain size analysis from XRD measurements (Fig. 4(b)) using the following Scherrer equation, $B_g = K\lambda/dcos\theta$ (K is the shape factor having a value of about 0.9, λ is the wavelength of Cu K-alpha x-ray at a value of

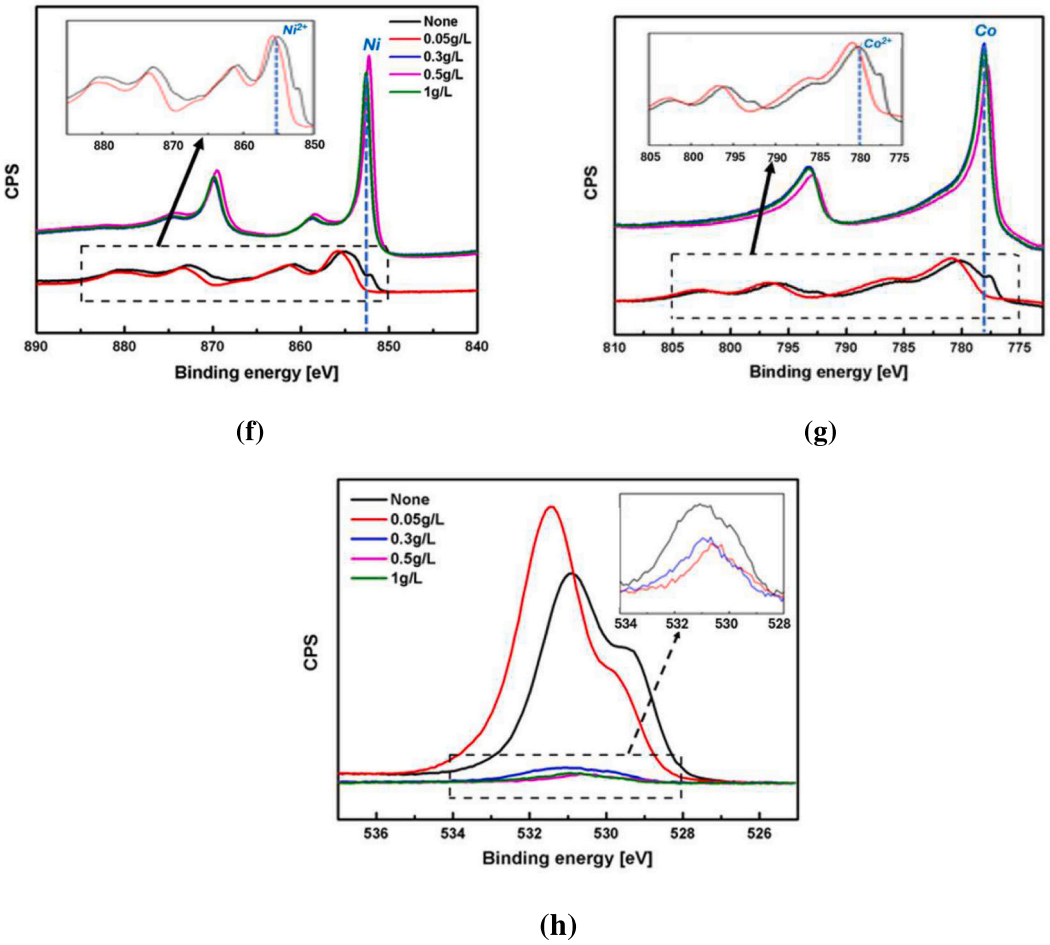


Fig. 3. (*continued*).

approximately 0.154 nm, θ the Bragg angle, and the crystal size of the material is presented by d). Specimens C, D, and E have smaller grain sizes compared to A and B as shown in Fig. 4(c), with specimens D and E having even smaller grains compared to C. However, specimen C exhibits the highest hardness value despite having larger grains compared to D and E, which can be attributed to the inverse Hall-Petch effect observed in nanocrystals with grain sizes near 10 nm or smaller, aligning with the grain sizes of specimens D and E [67–70]. Interestingly, the (220) plane is absent when saccharin concentrations exceed 0.3 g/L, highlighting saccharin's influence on determining the primary crystal orientation during growth. The underlying mechanism is briefly shown in Fig. A1.

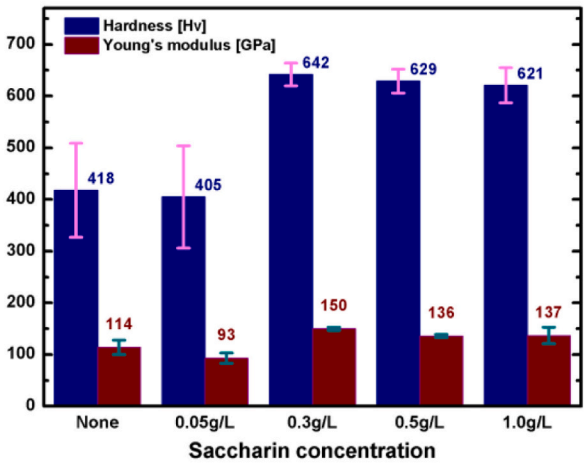
In MEMS probe tip fabrication, Young's modulus must remain relatively low to prevent brittle failure due to high stiffness. Although Young's modulus increases when saccharin addition is above 0.3 g/L, the values remain in the acceptable range under 150GPa. Typically, the grain boundaries are more vulnerable to plastic deformation compared to crystals, thus it is expected that Young's modulus decreases with finer grains [71]. However, the measured result shows that stiffness is higher in specimens C, D, and E, compared to specimens A and B. This is due to the compact, non-porous structures of films since pores in specimens A and B can be vulnerable to plastic deformation [72]. In summary, the addition of an optimal amount of saccharin can significantly enhance hardness by more than 50 %. However, excessive saccharin may slightly reduce hardness due to the inverse Hall-Petch effect while lowering Young's modulus, which can improve resistance to brittle failure.

The friction coefficient is essential for improved wear properties and durability. From micro scratch tests, the scratching coefficient of friction (SCOF) of the coating layer is obtained through the formula $SCOF = F_t/F_n$ [73], where F_n is the vertical load and F_t corresponds to the tangential load. Measurements were carried out by progressively increasing the vertical load from 30mN to 3000mN. As shown in Fig. 4 (d), all specimens exhibit lower friction coefficients across the entire range of vertical loads compared to specimen A, deposited without saccharin. While friction arises from various surface properties of materials, such as roughness, asperity deformation, adhesion, and plowing [74], friction due to plowing the specimen surface increases with hardness. Since specimens A and B have a much rougher surface compared to specimens C, D, and E, the high friction coefficient of A and B is attributed to surface roughness. However, when a lower range of load near 300mN is applied, specimen E has the lowest SCOF value, followed by D and C. As surface roughness is reduced in specimens C, D, and E, hardness becomes a key factor in determining the friction coefficient, with higher hardness correlating to a higher friction coefficient. Lastly, the scratch track, imaged using an optical microscope (Fig. 4(e)), reveals more pronounced deformation with a wider and rougher track in the deposit without saccharin (top) compared to the smoother, narrower track in the 0.5 g/L saccharin deposit (bottom), indicating lower friction coefficient in the presence of saccharin.

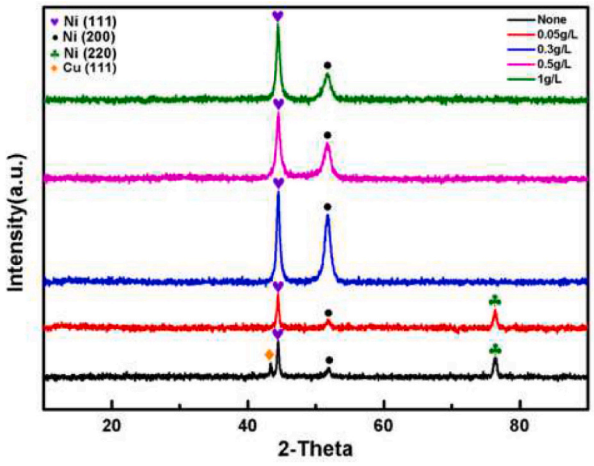
4. Discussion

4.1. Electrochemical impedance spectroscopy

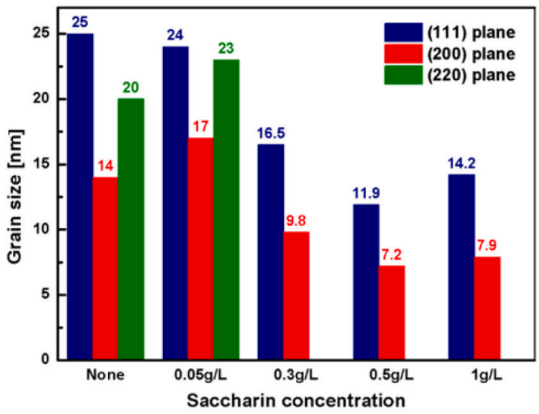
Electrochemical Impedance Spectroscopy (EIS) provides a unique advantage in analyzing the reaction kinetics at the metal-electrolyte interface. It characterizes the comprehensive impedance response arising from the complex interplay between the electrode and its



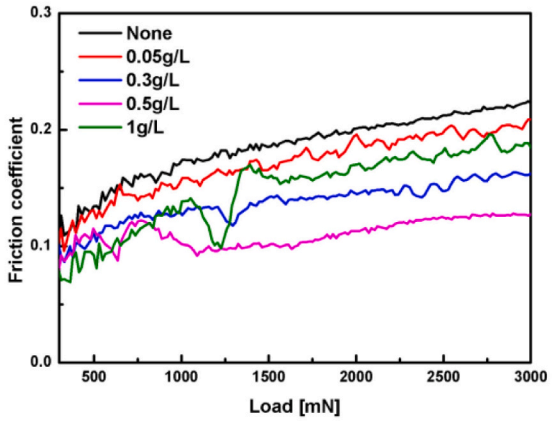
(a)



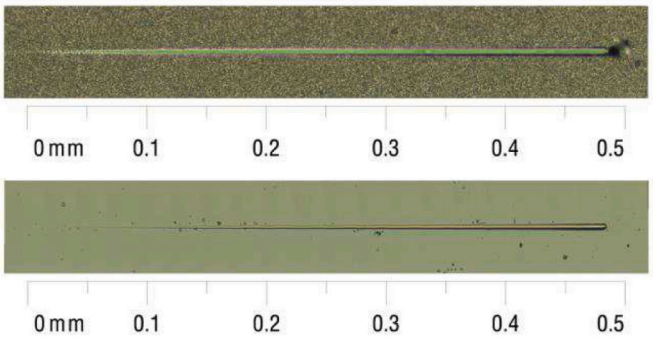
(b)



(c)



(d)



(e)

Fig. 4. (a) Vickers' hardness and Young's modulus values measured by nano-indentation. Measured values are the average of nine measurements and standard deviations are presented as error bars. (b) XRD patterns of Ni—Co films deposited from solutions with different saccharin concentrations. (c) The grain size of different crystal orientations in the deposited films, obtained from the XRD pattern. (d) Scratching Coefficient of Friction (SCOF) of Ni—Co films electrodeposited with various saccharin concentrations, measured using progressive load micro-scratch tests ranging from 300mN to 3000mN over 1 min. (e) Scratch-track images of deposits with and without 0.5 g/L saccharin, obtained from the progressive load micro-scratch test.

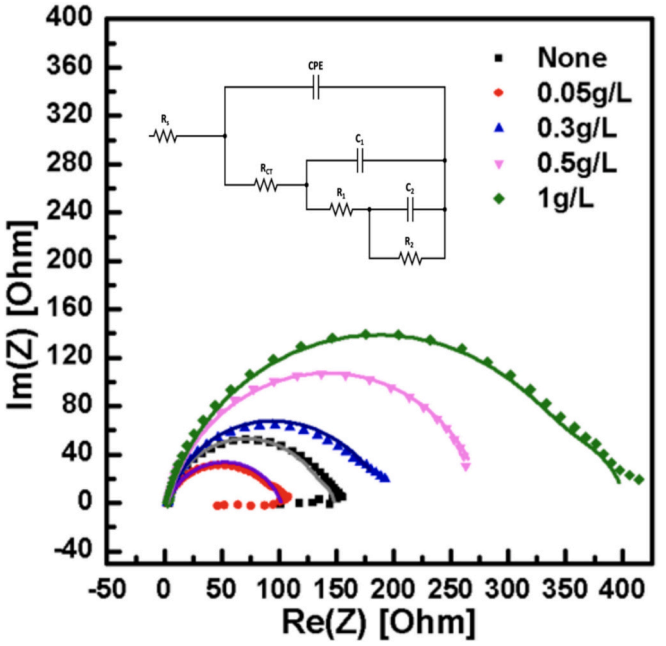


Fig. 5. Nyquist plots obtained from electrochemical impedance spectroscopy (EIS) at the nucleation overpotential with voltage perturbation of 5 mV. AC voltage frequency is swept from 150 kHz to 100mHz, corresponding to the dotted curve from left to right. Dotted data points are the measured data for different frequencies, and the real lines correspond to fitted curves using the equivalent circuit in the inset. Three time constants are used to model three charge transfer reactions involved in the electrodeposition (metal reduction, Volmer, and Heyrovsky reaction).

surrounding electrolyte. By probing impedance across a range of AC voltage frequencies, EIS reveals valuable insights into charge transfer phenomena and interfacial behavior at the molecular level.

Fig. 5 shows the Nyquist plot for the same overpotential near the nucleation potential at different saccharin concentrations. The impedance response shows compressed semi-circles in every saccharin concentration. Contrary to the high-frequency region, negative reactance in the low-frequency region is observed for low saccharin concentrations. Generally, the negative reactance in the low-frequency region is known to be caused by the adsorption of electrochemical species on the electrode surface [75,76]. There are mainly three species that can adsorb to the electrode surface during the electrodeposition process: H_{ads}, $M(OH)^{+}_{ads}$, and $M(OH)_2$, where M represents Ni or Co atoms. Contrary to the other species, $M(OH)_2$ forms and precipitates through the following chemical reaction, $M^{+} + 2OH^{-} \rightarrow M(OH)_2$, without involving a charge transfer process [77]. This film formation near the cathode during the electrodeposition is severe when the local pH rise is large. This process is expected to be responsible for the negative reactance in Fig. 5 at low concentrations. This will be dealt with in more detail in the later section through the analysis of the surface coverages of species.

The electrochemical process can be modeled by fitting to an equilibrium circuit. In this study, the reactions involving the metal reduction and hydrogen evolution reactions are fitted to an equilibrium circuit comprising three time constants because three charge transfer reactions are involved, namely the metal reduction, Volmer, and Heyrovsky reactions. The Tafel reaction can be neglected because it requires H_{ads} diffusion on the cathode which is much slower compared to the Volmer and Heyrovsky reactions in acidic baths [34]. This is because the proton diffusion in the acidic solution is much faster than substrate diffusion.

The equivalent circuit elements are interpreted as follows. CPE refers to a constant phase element that models the actual behavior of an electrochemical double layer, an imperfect capacitor [78]. The impedance of CPE is expressed by the equation $Z_{CPE} = 1/Y_0[(jw)^n]^{-1}$ (Y_0 is the

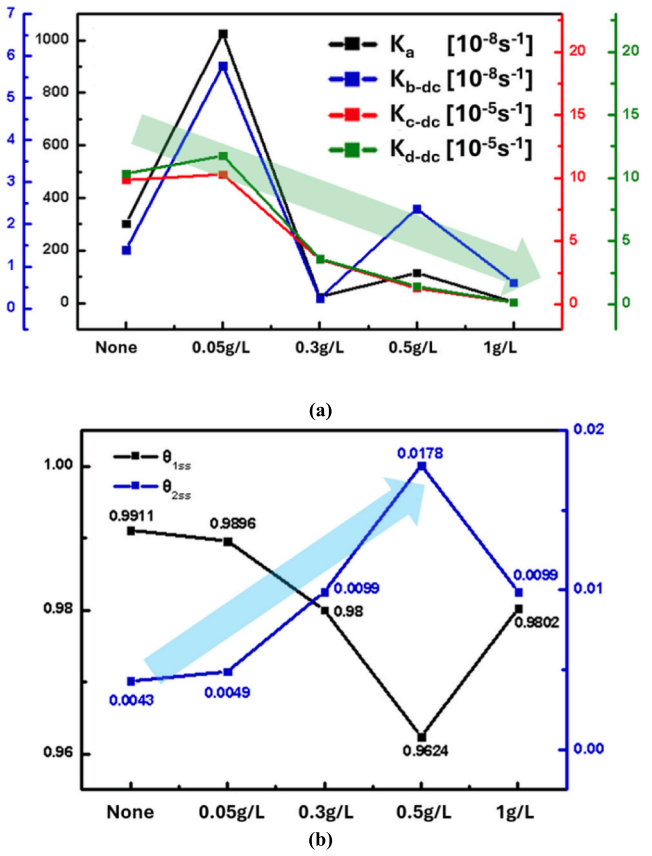


Fig. 6. (a) Steady-state effective reaction rate constants and (b) surface coverage values of $M(OH)^{+}$, H_{ads} obtained from the semi-empirical analysis obtained from the semi-empirical analysis in Appendix. Exact values are provided in Table A.1.

magnitude of CPE, j is the imaginary number, n has a value between 0.5 and 1, represented by the equation $n = \pi/(\alpha/2)$, and α as the phase angle of CPE). R_s is the solution resistance between the reference electrode and cathode. R_{CT} is the charge transfer resistance at the cathode surface [79]. C_1 is the pseudo-capacitance modeling the fast reversible reactions such as the adsorption-dissolution process of adsorbed intermediate species that behave like capacitors [80,81]. Lastly, C_2 is the capacitance relating to the adsorbed species. Each circuit parameter is obtained by fitting the experimental impedance with the EC-Lab Zfit software as shown in Table 3.

The obtained values of the circuit elements are consistent with the physical interpretation of the expected behavior of electrochemical species. The CPE value decreases with higher saccharin concentrations. This occurs because saccharin molecules near the cathode surface replace water molecules in the electric double layer when dissolved in the electroplating solution. This substitution reduces the double-layer capacitance, as the dipole moment of saccharin is lower than that of water molecules [82]. Additionally, the adsorbed saccharin increases the adsorption activation energy of H_3O^{+} species, further contributing to the reduction in the CPE value.

A higher charge transfer resistance (R_{CT}) is observed at higher saccharin concentrations, except for the case of 0.05 g/L, where R_{CT} decreases from 137.6 to 98.02 Ohm. The decreased R_{CT} at low concentrations is attributed to accelerated hydrogen reactions caused by reduced competition between metal and hydrogen adsorption. This reduced competition arises because adsorbed saccharin acts as a barrier for hydrogen ion adsorption, increasing the activation energy of hydrogen adsorption near saccharin molecules [83–86]. As a result, hydrogen reactions primarily occur at active sites with low saccharin adsorption while metal reduction occurs in the active sites near the

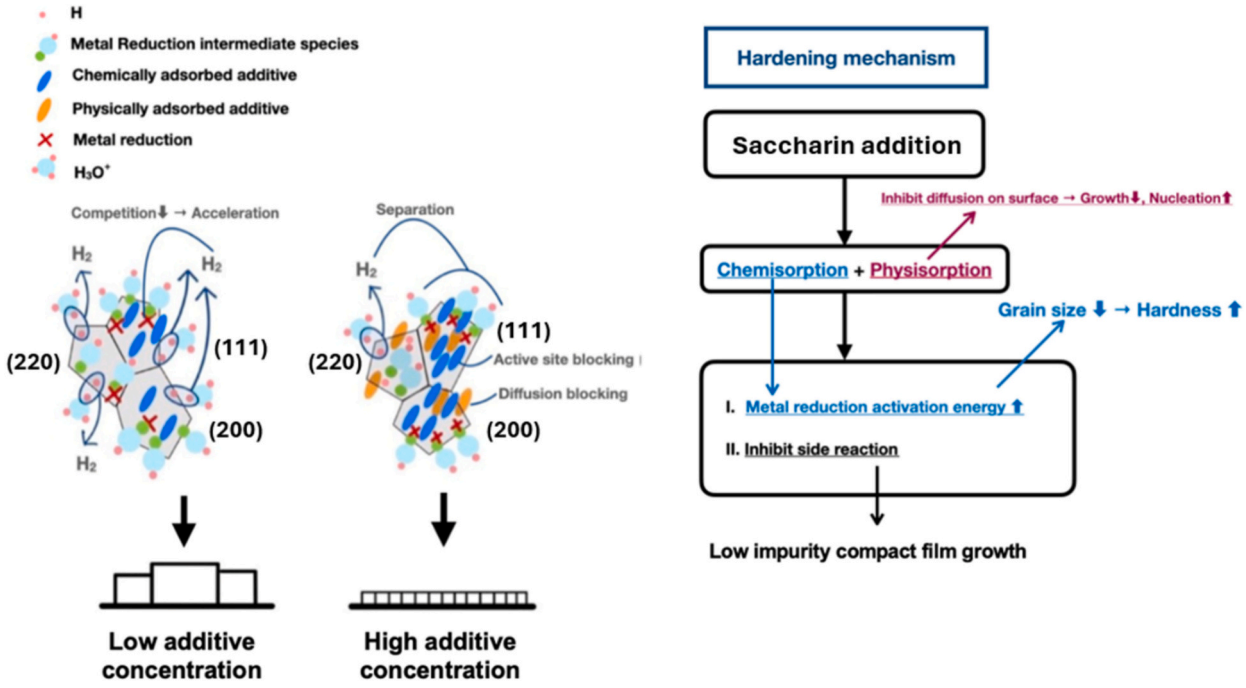


Fig. 7. Schematic of the substrate/electrolyte interface during nucleation and resulting microstructures at low and high saccharin concentrations. The hardening mechanism is summarized to the right of the schematic.

Table 2
Tafel equation, exchange current density, and charge transfer coefficient of the electrochemical reaction for Ni—Co alloy electrodeposition at different saccharin concentrations.

Saccharin concentration	Tafel equation	b (Slope) [mV/dec]	i_0 (Exchange current density) [μA/cm^2]	α
None	$\eta = 0.136\ logi + 0.57$	136	64.4	0.102
0.05 g/L	$\eta = 0.115\ logi + 0.63$	115	3.32	0.121
0.3 g/L	$\eta = 0.127\ logi + 0.68$	127	4.42	0.109
0.5 g/L	$\eta = 0.097\ logi + 0.72$	97	3.78×10^{-2}	0.144
1.0 g/L	$\eta = 0.058\ logi + 0.83$	58	4.89×10^{-9}	0.238

saccharin-dense region. This separation of reaction sites simultaneously accelerates metal and hydrogen reactions when saccharin coverage is low, as in the 0.05 g/L case where surface coverage is only 9 %. Conversely, at higher saccharin concentrations (above 0.3 g/L), the inhibitory effect of saccharin dominates, outweighing the acceleration mediated by site separation, as reflected in the higher R_{CT}.

4.2. Semi-empirical modeling of reaction kinetics using EIS data

The kinetic constants and steady-state surface coverage values are obtained through a semi-empirical analysis. These values are crucial to understanding the mechanism of the effect of additives during electrodeposition. The theoretical derivations are detailed in the Appendix. The main electrochemical reactions occurring at the electrode surface during Ni—Co electrodeposition are summarized in Eqs. 6–9, where *M* represents Ni and Co atoms, and *S* denotes the electrode surface. For convenience, the four reactions are labeled as *a, b, c*, and *d*. The surface coverage of $M(OH)^+_{ads}$ is denoted as θ_1, and the surface coverage of H_{ads} is θ_2. Reactions *a* and *c* are adsorption processes that occur on free active sites not occupied by any species (1- θ_1- θ_2). Conversely, reactions *b* and *d* are reduction processes that take place on adsorbed species, specifically on sites θ_1 and θ_2, respectively. The reaction rate constants are represented by k_a, k_b, k_c, and k_d.

(a)

$$M(OH)^+ \rightarrow M(OH)^+_{ads} \tag{6}$$

(b)

$$M(OH)^+_{ads} + 2e^- \rightarrow M + (OH)^- \tag{7}$$

(c)

$$H_3O^+ + e^- + S \rightarrow S - H_{ads} + H_2O \tag{8}$$

(d)

Table 3
Equivalent circuit element values fitted to the EIS data.

	R_s [Ohm]	CPE [F.s^{n-1}]	n	R_{CT} [Ohm]	C_1 [F]	R_1 [Ohm]	C_2 [F]	R_2 [Ohm]
None	2.475	$0.1217 \cdot 10^{-3}$	0.8369	137.6	$17.99 \cdot 10^{-3}$	2.823	$1.245 \cdot 10^{-3}$	5.286
0.05 g/L	2.5	$0.2412 \cdot 10^{-3}$	0.7593	98.02	$0.1005 \cdot 10^{-3}$	0.4317	$1.453 \cdot 10^{-3}$	1.098
0.3 g/L	2.5	$98.96 \cdot 10^{-6}$	0.814	179.6	$27.15 \cdot 10^{-6}$	2.353	$4.773 \cdot 10^{-3}$	8.055
0.5 g/L	2.509	$89.77 \cdot 10^{-6}$	0.8416	231.8	$3.114 \cdot 10^{-6}$	30.71	$0.27 \cdot 10^{-3}$	14.38
1 g/L	2.017	$82.26 \cdot 10^{-6}$	0.8242	357.1	$2.121 \cdot 10^{-6}$	10.2	$8.519 \cdot 10^{-3}$	32.09

$$S - H_{ads} + H_3O^+ + e^- \rightarrow S + H_{2(g)} + H_2O \quad (9)$$

K_a, $K_{b\text{-}dc}$, $K_{c\text{-}dc}$, and $K_{d\text{-}dc}$ represent the steady-state effective reaction rate constants for reactions *a, b, c*, and *d,* (see Eqs. A3, A9). The steady-state surface coverage values of $M(OH)^+_{ads}$ and H_{ads} are θ_{1ss} and θ_{2ss}, respectively. The calculated surface coverage values reveal that $M(OH)^+_{ads}$ species occupy 96 to 99 % of the active sites of the cathode depending on the saccharin concentration, while the surface coverage of adsorbed hydrogen is only 0.4 to 2 %. Despite the low surface coverage of hydrogen, the hydrogen reaction rates ($K_{c\text{-}dc}$ and $K_{d\text{-}dc}$) are significantly faster than the metal reduction reaction rates (K_a and $K_{b\text{-}dc}$). This allows the hydrogen adsorption to compete with metal adsorption during the nucleation process.

The apparent difference between the low and high saccharin concentrations is the reduced reaction rate of Volmer and Heyrovsky reactions ($K_{c\text{-}dc}$ and $K_{d\text{-}dc}$), where the suppression is at most 70 times. Contrary to the suppression of the Volmer reaction rate where hydrogen adsorption occurs, the surface coverage of adsorbed hydrogen (θ_{2ss}) increases over saccharin concentrations (except for the 1 g/L), as seen in the arrow in Fig. 6(b). The increase in the θ_{2ss} value is due to the decreased competition between the metal and hydrogen adsorption species due to the separation effect of saccharin.

The reaction rate calculation reveals the underlying reason for the decreased oxide/hydroxide impurities within the film for higher saccharin concentrations. There are two main reasons, first, it is because of the decreased precipitation of $M(OH)_2$ species. Here it is confirmed that the low-frequency negative reactance observed at the low saccharin concentrations in Fig. 5 is due to the $M(OH)_2$. The dominant species covering the active sites are $M(OH)^+_{ads}$ throughout all saccharin concentrations with minimal variations in θ_{1ss}. Additionally, θ_{2ss} is higher when saccharin concentration is high as opposed to the negative reactance observed in low concentrations. Therefore, the only species that can significantly alter the low-frequency reactance is the $M(OH)_2$. This is consistent with the reduced Volmer and Heyrovsky reaction rates in the high saccharin concentration since the $M(OH)_2$ precipitation increases in high local pH regions caused by rapid hydrogen consumption.

Second, the reduced delay between the two steps of the metal deposition is responsible for less impurities. For high saccharin concentrations, the ratio of metal reaction rates ($K_a/K_{b\text{-}dc}$) is significantly low, as the ratio decreases from 235 (None) to 5.7 (1 g/L) as saccharin concentration increases. This indicates that the rate of the metal adsorption and the reduction process become closer, reducing the delay from reaction *b*. Consequently, the incorporation of intermediate species during electrodeposition decreases due to the shortened delay in the two-step process.

In summary, the reaction kinetics are governed by the competing effects of saccharin: inhibition and separation. At low concentrations, separation dominates, while inhibition becomes stronger at higher concentrations. Inhibition occurs through active site blocking, whereas separation arises from reduced competition between metal and hydrogen adsorption. Despite having a general trend of decreasing reaction rates with increasing saccharin concentrations, a variation in 0.05 g/L and 0.5 g/L is observed as a peak in Fig. 6(a). At 0.05 g/L, the acceleration is attributed to the stronger separation effect as the surface coverage of saccharin is low. While at 0.5 g/L, the physically adsorbed saccharin increases which does not block the active sites but increases the activation energy of hydrogen adsorption. Therefore, a strong separation is induced which explains the slight acceleration peak of the metal two-step processes compared to 0.3 g/L in Fig. 6(a). At higher concentrations such as 1.0 g/L, the huge surface coverage leads to significant inhibition of reaction rates. Fig. 7 schematically illustrates the metal and hydrogen reduction processes in the presence of low and high saccharin concentrations. The competition between the inhibition and separation is highlighted along with the resulting grain size. Additionally, the dominant adsorption of saccharin in the denser crystal orientations determines the growth orientation of the film, following the mechanism outlined in Fig. A1. The associated hardening mechanism is also depicted.

5. Conclusions

In conclusion, the addition of saccharin to the electroplating solution of Ni—Co resulted in an altered surface morphology from rough and non-uniform to compact and uniform surfaces. This change in microstructure and grain size led to significant improvements in mechanical and tribological properties, rendering the films optimal for MEMS probe card applications. Specifically, the addition of 0.3 g/L of saccharin yielded remarkable enhancements, including a 54 % increase in hardness values (from 418 to 642 Hv) and a considerable reduction in the coefficient of friction, attributed to the uniform surface. XPS analysis confirmed significantly reduced oxide and hydroxide impurities in the films. Furthermore, EIS measurements provided insights into reaction kinetics, revealing predominance adsorption of Ni(OH) + and Co(OH) + species on the electrode surface, with a minimal surface coverage of hydrogen. However, despite its low surface coverage, rapid hydrogen reactions significantly impacted the microstructure. The final microstructure was determined by a competition between the separation and inhibition effects of saccharin.

This study extends previous research by demonstrating the benefits of saccharin addition during Ni—Co electrodeposition including significant improvements in mechanical and tribological properties, impurity reduction, and control over hydrogen evolution. By providing a deeper understanding of the complex role of additives, this work advances the knowledge of alloy electrodeposition processes and their optimization.

CRediT authorship contribution statement

Jae-Ik Choi: Writing – review & editing, Writing – original draft, Project administration, Formal analysis, Data curation, Conceptualization. **Inyeol Park:** Methodology, Data curation. **Suyeon Park:** Methodology, Data curation. **Min Jun Yi:** Methodology, Data curation. **Wonbong Jang:** Supervision, Project administration. **Jong Hyun Seo:** Supervision, Project administration.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Jong Hyun Seo reports financial support was provided by National Research Foundation of Korea. Jong Hyun Seo reports financial support was provided by Korea Ministry of Trade Industry and Energy. Jong Hyun Seo has patent #1-1-2022-1,175,476-43 issued to HANYANG International Patent And Law Firm (9–2000–100,005-4). If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Appendix

A.1. Theoretical calculation of reaction kinetics using EIS.

In this section, a semi-empirical analysis is performed by comparing the experimental circuit parameters of the equivalent circuit obtained from EIS measurements with the theoretical derivations based on the mass balance of the electrochemical reactions.

The surface coverage changes of $M(OH)^{+}_{ads}$ and H_{ads} species over time can be expressed by the following formula related to the mass balance equation [87]. K_j is the effective rate constant, the kinetic parameter related to the rate of reaction (k_j) occurring on the $1cm^2$ electrode surface.

$$\Gamma\frac{d\theta_1}{dt} = K_a(1-\theta_1-\theta_2) - K_b\theta_1 \tag{A.1}$$

$$\Gamma\frac{d\theta_2}{dt} = K_c(1-\theta_1-\theta_2) - K_d\theta_2 \tag{A.2}$$

$$K_j = \begin{cases} \Gamma_0(1-\theta_{saccharin})k_j \ (j=a,b) \\ \Gamma_0(1-\theta_{saccharin})[H_3O^+]k_j \ (j=c,d) \end{cases} \tag{A.3}$$

Γ_0 is the number of active sites on the electrode and generally has a value of about 10^{-8} mol/cm^2 [87–89]. $\theta_{saccharin}$ is the surface coverage of saccharin molecules and $\Gamma = \Gamma_0(1\text{-}\theta_{saccharin})$. At steady state ($d\theta/dt = 0$), the surface coverage θ_{1ss}, θ_{2ss} can be calculated as Eq. A.4. K denoted with dc refers to the values of K at a constant voltage.

$$\begin{pmatrix}\theta_{1ss}\\ \theta_{2ss}\end{pmatrix} = \frac{1}{K_aK_{d-dc} + K_{b-dc}K_{c-dc} + K_{b-dc}K_{d-dc}}\begin{pmatrix}K_aK_{d-dc}\\ K_{b-dc}K_{c-dc}\end{pmatrix} \tag{A.4}$$

Because three electrochemical reactions are accompanied, the Faradaic current can be expressed as Eq. A.5.

$$i_F = 2FK_b\theta_1 + FK_c(1-\theta_1-\theta_2) + FK_d\theta_2 \tag{A.5}$$

The reaction rate constants and surface coverage values of $M(OH)^{+}_{ads}$, and H_{ads} change in varying electrode potential. Since the potential of the AC component (voltage perturbation in the EIS measurements) can be considered small, changing variables (θ_i, k_j, and K_j) over E_{ac} can be approximated by Taylor series expansion (to the first order) as shown in Eq. A.6, A.7 [87]. Rate constant (k_j) is dependent on the electrode potential because the charge transfer is involved. Therefore, k_j can be expressed by the Tafel law as presented in Eq. A.7 [87,89,90]. $K_{j\text{-}dc}$ are also presented as Eq. A.9. E_{dc}, E_{ac} are DC and AC components of the electrode potential, respectively. k_{j0} is the standard rate constant that is a measure of the rate at which the system approaches equilibrium, b_j (j = b,c,d) refers to the Tafel constant of reaction *b, c,* and *d*.

$$\theta_i = \theta_{iss} + \frac{\Delta\theta_i}{\Delta E}E_{ac} \ (i=1,2) \tag{A.6}$$

$$k_j = k_{j0}exp(b_jE) = k_{j0}exp\{b_j(E_{dc}+E_{ac})\} = k_{j-dc}(1+b_jE_{ac}) \ (j=b,c,d) \tag{A.7}$$

$$k_{j-dc} = k_{j0}exp(b_jE_{dc}) \ (j=b,c,d) \tag{A.8}$$

$$K_{j-dc} = \begin{cases} \Gamma_0(1-\theta_{saccharin})k_{j-dc} \ (j=a,b) \\ \Gamma_0(1-\theta_{saccharin})[H_3O^+]k_{j-dc} \ (j=c,d) \end{cases} \tag{A.9}$$

E_{ac} can be expressed in the form of a sinusoidal wave having a magnitude of E^0_{ac}, and frequency ω as Eq. A.10.

$$E_{ac} = E^0_{ac}\,sin(\omega t) \tag{A.10}$$

Thus, the change of potential over time is expressed as Eq. A.11.

$$\frac{\Delta E}{\Delta t} = \frac{\Delta(E_{dc}+E_{ac})}{\Delta t} = \frac{\Delta E_{ac}}{\Delta t} = E^0_{ac}\omega cos(\omega t) = E^0_{ac}\omega sin\left(\omega t + \frac{\pi}{2}\right) = j\omega E^0_{ac}sin(\omega t) = j\omega E_{ac} \tag{A.11}$$

$$\Gamma \bullet \frac{\Delta\theta_i}{\Delta t} = \Gamma \bullet \frac{\Delta E}{\Delta t} \bullet \frac{\Delta\theta_i}{\Delta E} \ (i=1,2) \tag{A.12}$$

When substituting ΔE/Δt in Eq. A.11 to Eq. A.12, and substituting the obtained $\Delta\theta_i/\Delta t$ with Eq. A.1 and A.2 (the mass balance equations) and developing θ_i and K_j by the Taylor series expansion as Eq. A.6, A.7, the following matrix of the surface coverage change over voltage can be obtained.

$$\begin{pmatrix} j\omega\Gamma + K_a + K_{b-dc} & K_a \\ K_{c-dc} & j\omega\Gamma + K_{c-dc} + K_{d-dc}\end{pmatrix}\begin{pmatrix}\frac{\Delta\theta_1}{\Delta E}\\ \frac{\Delta\theta_2}{\Delta E}\end{pmatrix} = \begin{pmatrix} b_bK_{b-dc}\theta_{1ss} \\ b_cK_{c-dc}(1-\theta_{1ss}-\theta_{2ss}) - b_dK_{d-dc}\theta_{2ss}\end{pmatrix} \tag{A.13}$$

The Faradaic current i_F depends on three variables E, θ_1, θ_2. Accordingly, the amount of i_F change over sufficiently small AC voltage perturbation can be expressed as follows.

$$\frac{\Delta i_F}{\Delta E} = \left(\frac{\partial i_F}{\partial E}\right)_{\theta_1,\theta_2} + \left(\frac{\partial i_F}{\partial\theta_1}\right)_{E,\theta_2} \bullet \frac{\Delta\theta_1}{\Delta E} + \left(\frac{\partial i_F}{\partial\theta_2}\right)_{E,\theta_1} \bullet \frac{\Delta\theta_2}{\Delta E} \tag{A.14}$$

Differentiating the following Eq. A.5, Eq. A.15 and A.16 are obtained. Moreover, partially differentiating Eq. A.5 over the voltage when θ_1 and θ_2 are constants, Eq. 17 is obtained. Eq. A.18 is the defined Faradaic admittance. Next, by substituting Eq. A.15, A.16, and A.17 in Eq. A.14, Eq. A.19 is obtained. Afterward, by substituting $\Delta\theta_1/\Delta E$, $\Delta\theta_2/\Delta E$ in Eq. A.19 with Eq. A.13, the formula of Faradaic admittance in Eq. A.20 is obtained. The real part of the formula is expressed as *Real* for simplicity because it is not used in comparing coefficients with the Faradaic admittance expression obtained

from the circuit.

$$\left(\frac{\partial i_F}{\partial \theta_1}\right)_{E,\theta_2} = F(2K_{b-dc} - K_{c-dc}) \tag{A.15}$$

$$\left(\frac{\partial i_F}{\partial \theta_2}\right)_{E,\theta_1} = F(K_{d-dc} - K_{c-dc}) \tag{A.16}$$

$$\left(\frac{\partial i_F}{\partial E}\right)_{\theta_1,\theta_2} = \frac{1}{R_{CT}} \tag{A.17}$$

$$\frac{\Delta i_F}{\Delta E} = \frac{1}{Z_f} \tag{A.18}$$

$$\frac{1}{Z_f} = \frac{1}{R_{CT}} + F(2K_{b-dc} - K_{c-dc}) \bullet \frac{\Delta\theta_1}{\Delta E} + F(K_{d-dc} - K_{c-dc}) \bullet \frac{\Delta\theta_2}{\Delta E} \tag{A.19}$$

$$\frac{1}{Z_f} = \frac{1}{R_{CT}} + \frac{(j\omega)\frac{F}{\Gamma}[-(2K_{b-dc} - K_{c-dc})K_{b-dc}b_b\theta_{1ss} + (K_{c-dc} - K_{d-dc})\{b_cK_{c-dc}(1-\theta_{1ss}-\theta_{2ss}) - b_dK_{d-dc}\theta_{2ss}\}] + Real}{-\omega^2 + \frac{1}{\Gamma}(K_a + K_{b-dc} + K_{c-dc} + K_{d-dc})j\omega + \frac{1}{\Gamma^2}(K_aK_{d-dc} + K_{b-dc}K_{c-dc} + K_{b-dc}K_{d-dc})} \tag{A.20}$$

The following calculations are performed to derive the Faradaic admittance from the equivalent circuit. Developing Eq. A.5 by Taylor series expansion, Eq. A.21 is obtained. Also, Faradaic impedance in this case is represented by a formula of circuit elements expressed in Eq. A.22 [87]. Where Z_a is defined in Eq. A.23. Since the admittance is expressed as the reciprocal of impedance, the Faradaic admittance can be expressed by the following formula in Eq. A.24.

$$\left(\frac{\partial i_F}{\partial E}\right)_{\theta_1,\theta_2} = \frac{1}{R_{CT}} = F\{2b_bK_{b-dc}\theta_{1ss} + b_cK_{c-dc}(1-\theta_{1ss}-\theta_{2ss}) + b_dK_{d-dc}\theta_{2ss}\} \tag{A.21}$$

$$Z_f = R_{CT} + \frac{1}{jwC_1 + \frac{1}{Z_a}} \tag{A.22}$$

$$Z_a = R_1 + \frac{1}{jwC_2 + \frac{1}{R_2}} \tag{A.23}$$

$$\frac{1}{Z_f} = \frac{1}{R_{CT}} + \frac{-jw\frac{C_2}{R_{CT}^2} - \frac{1}{R_{CT}^2}\left(\frac{1}{R_1} + \frac{1}{R_2}\right)}{-w^2 + jw\left(\frac{1}{C_2}\left(\frac{1}{R_1} + \frac{1}{R_2}\right) + \frac{1}{C_1}\left(\frac{1}{R_1} + \frac{1}{R_{CT}}\right) + \frac{1}{C_1C_2}\left(\frac{1}{R_1R_{CT}} + \frac{1}{R_2R_{CT}} + \frac{1}{R_1R_2}\right)\right)} \tag{A.24}$$

Finally, by equating coefficients from Eq. A.20 (Faradaic admittance obtained from the mass balance of electrochemical species in the chemical reactions) and Eq. A.24, the series of equations in Eq. A.25 is obtained.

$$\begin{cases} \dfrac{K_{b-dc}K_{d-dc}\{-b_b(2K_{b-dc} - K_{c-dc})K_a + (K_{c-dc} - K_{d-dc})(b_c - b_d)K_{c-dc}\}}{K_aK_{d-dc} + K_{b-dc}K_{c-dc} + K_{b-dc}K_{d-dc}} = -\dfrac{C_2}{R_{CT}^2} \bullet \dfrac{\Gamma}{F} \\ \dfrac{K_{b-dc}K_{d-dc}\{2b_bK_a + (b_c + b_d)K_{c-dc}\}}{K_aK_{d-dc} + K_{b-dc}K_{c-dc} + K_{b-dc}K_{d-dc}} = \dfrac{1}{R_{CT}} \bullet \dfrac{1}{F} \\ K_a + K_{b-dc} + K_{c-dc} + K_{d-dc} = \Gamma\left\{\dfrac{1}{C_2}\left(\dfrac{1}{R_1} + \dfrac{1}{R_2}\right) + \dfrac{1}{C_1}\left(\dfrac{1}{R_1} + \dfrac{1}{R_{CT}}\right)\right\} \\ K_aK_{d-dc} + K_{b-dc}K_{c-dc} + K_{b-dc}K_{d-dc} = \Gamma^2\dfrac{1}{C_1C_2}\left(\dfrac{1}{R_1R_{CT}} + \dfrac{1}{R_2R_{CT}} + \dfrac{1}{R_1R_2}\right) \end{cases} \tag{A.25}$$

By substituting the circuit element values from Table 3 and the Tafel slopes (b_j ($j = b,c,d$)) from Table 2 into the series of equations, the effective rate constants, and surface coverage of $M(OH)^+_{ads}$ and H_{ads} species are calculated. The obtained values are presented in Table A1.

Table A1
Steady-state kinetic parameters of the electrochemical reactions during Ni—Co electrodeposition, and surface coverage values of $M(OH)^+_{ads}$, H^+_{ads}, obtained by solving the series of equations in Eq. A.25.

	K_a [$10^{-8} \cdot s^{-1}$]	$K_{b\text{-}dc}$ [$10^{-8} \cdot s^{-1}$]	$K_{c\text{-}dc}$ [$10^{-8} \cdot s^{-1}$]	$K_{d\text{-}dc}$ [$10^{-8} \cdot s^{-1}$]	θ_{1ss}	θ_{2ss}
None	**303**	**1.29**	**$9.84 \cdot 10^3$**	**$10.36 \cdot 10^3$**	**0.9911**	**0.0043**
0.05 g/L	**$1.03 \cdot 10^3$**	**5.77**	**$10.30 \cdot 10^3$**	**$11.77 \cdot 10^3$**	**0.9896**	**0.0049**
0.3 g/L	**23.73**	**0.24**	**$3.56 \cdot 10^3$**	**$3.60 \cdot 10^3$**	**0.9800**	**0.0099**
0.5 g/L	**33.02**	**0.70**	**$1.34 \cdot 10^3$**	**$1.38 \cdot 10^3$**	**0.9600**	**0.0197**
1 g/L	**3.58**	**0.62**	**139.11**	**141.69**	**0.9802**	**0.0099**

A.2. Modification of Ni—Co growth plane orientation from saccharin.

The preferential growth orientation of Ni—Co shifts in the presence of saccharin, as confirmed by the XRD patterns in Fig. 4(b). Notably, the (220)

plane disappears when the saccharin concentration exceeds 0.3 g/L. This behavior is attributed to saccharin's role in suppressing hydrogen evolution. In regions with high saccharin adsorption, hydrogen reactions are more effectively suppressed, reducing competition between metal and hydrogen species. Consequently, metal species adsorption and reduction become more pronounced near saccharin-rich areas.

Since saccharin preferentially adsorbs onto denser Ni—Co planes, particularly (111) and (200), through coordination bonding with the metal, these planes become the dominant growth directions, rendering (220) less favorable. As a result, Ni—Co growth occurs predominantly at higher-energy (111) and (200) sites, increasing nucleation overpotential and promoting finer grain formation [58,59]. A schematic summarizing the analysis of growth orientation is presented in Fig. A1.

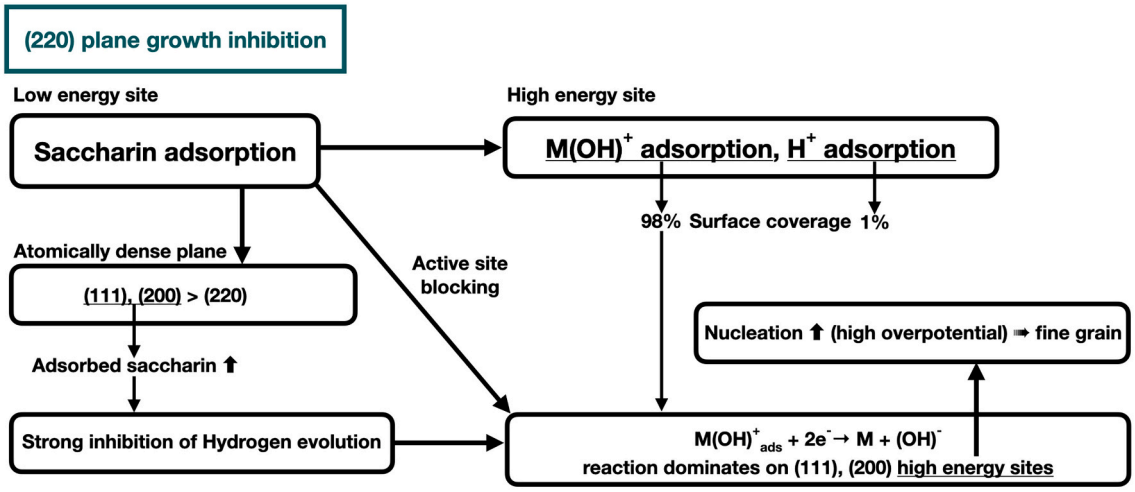


Fig. A1. Effect of saccharin adsorption on plane orientation development during the electrodeposition. The preferential adsorption of saccharin on atomically dense planes, such as (111) and (200), inhibits the growth of the (220) plane at high saccharin concentrations. The schematic illustrates the process flow, emphasizing the growth on dense planes. The measured plane orientation from the XRD is shown in Fig. 4 (b).

Data availability

Data will be made available on request.

References

[1] Y.-J. Chang, C.-Y. Ho, C.-H. Chang, Mater. Sci. Eng. C 37 (2014) 369.
[2] Y.-J. Chang, C.-H. Chang, J. Lab. Autom. 21 (2016) 394.
[3] W.-C. Choi, J.-Y. Ryu, Microsyst. Technol. 18 (2012) 333.
[4] J.Y. Kim, H.J. Lee, Y.-H. Cho, Microelectron. Eng. 87 (2010) 1769.
[5] C. Xu, Y. Xu, X. Gong, 2010 3rd International Nanoelectronics Conference (INEC), 2010, p. 1072.
[6] M. Bhuiyan, H. Quadri, M. Bhuiyan, M. Kajihara, J. Mater. Sci. Mater. Electron. 24 (2013) 3175.
[7] K. Liu, M. Chen, C. Pan, M. Chang, W. Huang, Sensors Actuators A Phys. 159 (2010) 126.
[8] B. Chi, J. Li, X. Yang, Y. Gong, N. Wang, Int. J. Hydrog. Energy 30 (2005) 29.
[9] G. Wu, N. Li, D. Zhou, K. Mitsuo, Surf. Coat. Technol. 176 (2004) 157.
[10] A. Sharma, S. Chhangani, R. Madhavan, S. Suwas, J. Magn. Magn. Mater. 434 (2017) 68.
[11] G. Qiao, T. Jing, N. Wang, Y. Gao, X. Zhao, J. Zhou, W. Wang, Electrochim. Acta 51 (2005) 85.
[12] L. Tian, J. Xu, S. Xiao, Vacuum 86 (2011) 27.
[13] L. Wang, Y. Gao, Q. Xue, H. Liu, T. Xu, Appl. Surf. Sci. 242 (2005) 326.
[14] M. Srivastava, V.E. Selvi, V.W. Grips, K. Rajam, Surf. Coat. Technol. 201 (2006) 3051.
[15] L. Qin, J. Lian, Q. Jiang, J. Alloys Compd. 504 (2010) S439.
[16] M. Zamani, A. Amadeh, S.L. Baghal, Trans. Nonferrous Metals Soc. China 26 (2016) 484.
[17] B. Tian, Y. Cheng, Electrochim. Acta 53 (2007) 511.
[18] X. Yi, P. He, Y. Chen, W. Wang, D. Yang, F. Dong, ECS Trans. 28 (2010) 13.
[19] V. Torabinejad, M. Aliofkhazraei, A.S. Rouhaghdam, M. Allahyarzadeh, Tribol. Int. 106 (2017) 34.
[20] A.R. Shetty, A.C. Hegde, Surf. Coat. Technol. 322 (2017) 99.
[21] Y. Yang, Y. Cheng, Surf. Coat. Technol. 216 (2013) 282.
[22] A. Karimzadeh, M. Aliofkhazraei, F.C. Walsh, Surf. Coat. Technol. 372 (2019) 463.
[23] L. Tian, J. Xu, C. Qiang, Appl. Surf. Sci. 257 (2011) 4689.
[24] M. Farzaneh, M. Zamanzad-Ghavidel, K. Raeissi, M. Golozar, A. Saatchi, S. Kabi, Appl. Surf. Sci. 257 (2011) 5919.
[25] L. Wang, Y. Gao, Q. Xue, H. Liu, T. Xu, Appl. Surf. Sci. 242 (2005) 326.
[26] I. Khazi, U. Mescheder, Materials Research Express 6 (2019) 082001.
[27] J. Gu, B. Liu, G. Chen, X. Li, J. Micromech. Microeng. 26 (2016) 075009.
[28] N. Zech, E. Podlaha, D. Landolt, J. Electrochem. Soc. 146 (1999) 2886.
[29] H. Dahms, I. Croll, J. Electrochem. Soc. 112 (1965) 771.
[30] J. Park, T. Kwon, J. Kim, H. Jin, H.Y. Kim, B. Kim, S.H. Joo, K. Lee, Chem. Soc. Rev. 47 (2018) 8173.
[31] L. Tian, J. Xu, C. Qiang, Appl. Surf. Sci. 257 (2011) 4689.
[32] S. Gadad, T.M. Harris, J. Electrochem. Soc. 145 (1998) 3699.
[33] J. Santos, F. Trivinho-Strixino, E. Pereira, Surf. Coat. Technol. 205 (2010) 2585.
[34] T. Shinagawa, A.T. Garcia-Esparza, K. Takanabe, Sci. Rep. 5 (2015) 1.
[35] A. Lasia, Int. J. Hydrog. Energy 44 (2019) 19484.
[36] L. Mattarozzi, S. Cattarin, N. Comisso, A. Gambirasi, P. Guerriero, M. Musiani, L. Vázquez-Gómez, E. Verlato, Electrochim. Acta 140 (2014) 337.
[37] L.B. Harris, J. Electrochem. Soc. 120 (1973) 1034.
[38] W. Wu, N. Eliaz, E. Gileadi, Thin Solid Films 616 (2016) 828–837.
[39] W. Wu, Applied Physics A 122 (12) (2016) 1028.
[40] Z. Jamil, E. Ruiz-Trejo, N.P. Brandon, J. Electrochem. Soc. 164 (2017) D210.
[41] H.F. Alesary, S. Cihangir, A.D. Ballantyne, R.C. Harris, D.P. Weston, A.P. Abbott, K. S. Ryder, Electrochim. Acta 304 (2019) 118.
[42] A.B. Hashemi, G. Kasiri, F. La Mantia, Electrochim. Acta 258 (2017) 703.
[43] Y.-L. Zhu, Y. Katayama, T. Miura, Electrochim. Acta 85 (2012) 622.
[44] R. Oriňáková, A. Turoňová, D. Kladeková, M. Gálová, R.M. Smith, J. Appl. Electrochem. 36 (2006) 957.
[45] D. Golodnitsky, N. Gudin, G. Volyanuk, Plat. Surf. Finish. 85 (1998) 65.
[46] Y. Li, Y. Zhang, C. Zhang, Surf. Coat. Technol. 202 (2008) 4952.
[47] R. Riastuti, S.T. Siallagan, A. Rifki, F. Herdino, C. Ramadini, IOP Conf 541. 1. (2019).
[48] Y.-L. Zhu, Y. Katayama, T. Miura, Electrochim. Acta 123 (2014) 303.
[49] X. Cui, W. Chen, J. Electrochem. Soc. 155 (2008) K133.
[50] G.F. Wang, K.C. Chan, K.F. Zhang, Scr. Mater. 54 (2006) 765.
[51] R. Mishra, R. Balasubramaniam, Corros. Sci. 46 (2004) 3019.
[52] A. Kotelnikova, T. Zubar, T. Vershinina, M. Panasyuk, O. Kanafyev, V. Fedkin, A. Trukhanov, RSC Adv. 12 (55) (2022) 35722–35729.
[53] S. Budi, R.A. Tawwabin, U. Cahyana, M. Paristiowati, Int. J. Electrochem. Sci. 15 (7) (2020) 6682–6694.
[54] I.M. Omar, K.M. Emran, M. Aziz, A.M. Al-Fakih, RSC Adv. 10 (2020) 32113.
[55] Y. Yang, H. Yang, C. Liang, DEStech Trans. Mater. Sci. Eng. (amst) (2016).
[56] A.J. Bard, L.R. Faulkner, H.S. White, Electrochemical Methods: Fundamentals and Applications, John Wiley & Sons, 2022, pp. 88–100.
[57] J. Mostany, B.R. Scharifker, K. Saavedra, C. Borrás, Russ. J. Electrochem. 44 (2008) 652.
[58] J. Molina, B.A. Hoyos, Electrochim. Acta 54 (6) (2009) 1784–1790.
[59] A.M. Rashidi, A. Amadeh, Surf. Coat. Technol. 202 (16) (2008) 3772–3776.
[60] A.M. Al-Fakih, H.H. Abdallah, M. Aziz, Mater. Corros. 70 (2019) 135.
[61] A.P. Grosvenor, M.C. Biesinger, R.S.C. Smart, N.S. McIntyre, Surf. Sci. 600 (2006) 1771.
[62] M. Al-Kuhaili, S. Ahmad, S. Durrani, M. Faiz, A. Ul-Hamid, Mater. Sci. Semicond. Process. 39 (2015) 84.
[63] J.-K. Kang, S.-W. Rhee, Thin Solid Films 391 (2001) 57.
[64] J. Yang, H. Liu, W.N. Martens, R.L. Frost, J. Phys. Chem. C 114 (2010) 111.

[65] J. Chastain, R.C. King Jr., Perkin-Elmer Corporation 40 (1992) 221.
[66] S. Zak, C. Trost, P. Kreiml, M. Cordill, J. Mater. Res. 37 (2022) 1373.
[67] B. Wu, C. Schuh, P. Ferreira, Metall. Mater. Trans. A 36 (2005) 1927.
[68] K. Kadau, P. Lomdahl, B. Holian, T. Germann, D. Kadau, P. Entel, D. Wolf, M. Kreth, F. Westerhoff, Metall. Mater. Trans. A 35 (2004) 2719.
[69] C. Carlton, P. Ferreira, Acta Mater. 55 (2007) 3749.
[70] X. Liu, F. Yuan, Y. Wei, Appl. Surf. Sci. 279 (2013) 159.
[71] M. Becton, X. Wang, Phys. Chem. Chem. Phys. 17 (2015) 21894.
[72] R.A. Hardin, C. Beckermann, Metall. Mater. Trans. A 38 (2007) 2992.
[73] G. Subhash, W. Zhang, Wear 252 (2002) 123.
[74] K. Lee, K.P. Marimuthu, C.-L. Kim, H. Lee, Tribol. Int. 120 (2018) 398.
[75] S. Tebbakh, Y. Messaoudi, A. Azizi, N. Fenineche, G. Schmerber, A. Dinia, Trans. IMF 93 (2015) 196.
[76] M. Ahmadi, M.J. Guinel, J. Alloys Compd. 574 (2013) 196.
[77] D.S. Hall, D.J. Lockwood, C. Bock, B.R. MacDougall, Proceedings of the Royal Society a: mathematical, Phys. Eng. Sci. 471 (2015) 20140792.
[78] M.A. Amin, J. Appl. Electrochem. 36 (2006) 215.
[79] M. Gholami, I. Danaee, M.H. Maddahy, M. Rashvandavei, Ind. Eng. Chem. Res. 52 (2013) 14875.
[80] M. Schalenbach, Y.E. Durmus, H. Tempel, H. Kungl, R.-A. Eichel, Sci. Rep. 12 (2022) 1.
[81] B.E. Conway, E. Gileadi, Trans. Faraday Soc. 58 (1962) 2493.
[82] S. Hassani, K. Raeissi, M. Golozar, J. Appl. Electrochem. 38 (2008) 689.
[83] S.H. Ahn, S.J. Hwang, S.J. Yoo, I. Choi, H.-J. Kim, J.H. Jang, S.W. Nam, T.-H. Lim, T. Lim, S.-K. Kim, J. Mater. Chem. 22 (2012) 15153.
[84] O. Beeck, Rev. Mod. Phys. 17 (1945) 61.
[85] O. Beeck, A. Ritchie, Discuss. Faraday Soc. 8 (1950) 159.
[86] C. Ma, S. Wang, C. Low, L. Wang, F. Walsh, Trans. IMF 92 (2014) 189.
[87] D.-S. Kong, Y.-Y. Feng, J. Electrochem. Soc. 156 (2009) C283.
[88] D. Vanmaekelbergh, B. Erné, J. Electrochem. Soc. 146 (1999) 2488.
[89] D. Giménez-Romero, J. Garcia-Jareno, F. Vicente, Electrochem. Commun. 5 (2003) 722.
[90] L. Bai, B. Conway, J. Electrochem. Soc. 138 (1991) 2897.